Infosys Springboard Virtual Internship 6.0 Completion Report

Batch Number: **13**

Start date:**02-Feb-2026**

Name: **Shweta Nagraj Purvi**

Internship Duration: **8 Weeks**

1. Project Title

NeuroFleetX – AI-Driven Urban Mobility Optimization System

2. Project Objective

The primary objective of this project is to design and develop an intelligent urban mobility system that enhances transportation efficiency using AI-driven decision-making. The system aims to reduce traffic congestion, optimize vehicle utilization, and improve user experience through smart booking, real-time tracking, and predictive analytics.

3. Project description in detail

NeuroFleetX is a next-generation AI-driven platform designed to optimize urban mobility and fleet operations for rental, transport, and smart city use cases. By leveraging artificial intelligence, IoT, and geospatial data, NeuroFleetX enables real-time vehicle tracking, intelligent routing, predictive maintenance, dynamic fleet allocation, and customer-centric service models.

The system integrates machine learning models to predict traffic conditions, optimize route plans, and monitor vehicle health. It also employs advanced dashboards and mobile-first interfaces for fleet managers and end-users. Designed for scalability, NeuroFleetX can be adapted for electric vehicle (EV) fleets, ride-sharing models, or multi-modal transport networks, offering both operational efficiency and sustainable mobility solutions for modern cities.

NeuroFleetX is an AI-powered fleet management and urban mobility platform designed to optimize transportation systems in modern cities. The system integrates real-time tracking, intelligent routing, and predictive analytics to improve operational efficiency.

The platform supports multiple user roles such as Customer, Driver, and Fleet Manager, each with dedicated functionalities. It includes modules for booking vehicles, monitoring trips, managing user identity, and analyzing system performance through dashboards.

Advanced features such as AI Model Insights and Signal Center provide real-time alerts and decision support. A chatbot system is also included to assist users with queries and navigation.

The system is scalable and can be extended to support electric vehicles, ride-sharing services, and smart city infrastructure, contributing to sustainable and efficient mobility solutions.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project planning, requirement analysis	Defined system architecture and user roles (Customer, Driver, Fleet Manager)
Week 2	UI/UX design	Designed wireframes and layouts for dashboards and booking flow
Week 3	Frontend setup (React + Vite + MUI)	Initialized project, built basic components and routing
Week 4	Authentication & backend integration.	Implemented login/signup and role-based access control
Week 5	Dashboard & Book Vehicle module	Built dashboard and booking functionality.
Week 6	Core modules development	Developed Home, Trips, Signals, Identity, Settings
Week 7	AI features & system integration	Integrated AI insights and connected modules
Week 8	Testing, optimization & final enhancements	Fixed bugs, improved UI, added map & alerts

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Finalized idea, defined scope, and planned system architecture	Week 1
Prototype/First Draft	Initial UI design and frontend setup completed	Week 3
Mid-Term Review	Core features like authentication and booking system implemented	Week 5
Final Submission	Full system integration with all dashboards and features completed	Week 8
Presentation	Demonstration of working system and explanation of architecture	Week 8

5b. Project execution details

The project was executed in a structured and phased manner, beginning with requirement analysis and system design. Authentication modules such as login and registration were developed first with secure backend integration.

Core modules including Dashboard and Book Vehicle were implemented using a component-based architecture in React, ensuring modular and maintainable code. Additional pages like Home, Signals, My Trips, Identity, and Settings were developed and integrated using proper routing and responsive design.

AI-based features were incorporated through the AI Model Insight module, which analyzes traffic data and system parameters to generate actionable insights. These insights can be approved or dismissed and are stored in the database.

The system also includes a map-based booking module that allows users to select pickup and drop locations, view available vehicles, and receive optimized route suggestions. Final

improvements included UI enhancements, validation, testing, and performance optimization.

6. Snapshots / Screenshots

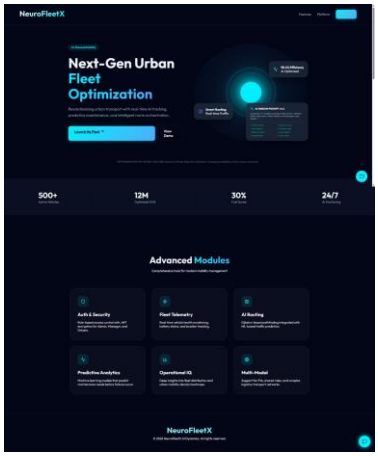

FIG 1. LANDING PAGE (HOME SCREEN)

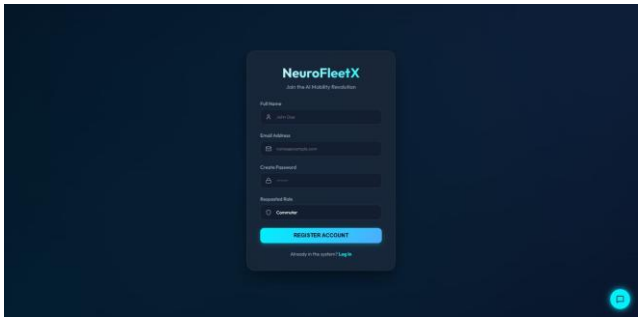

FIG 2. USER REGISTRATION INTERFACE

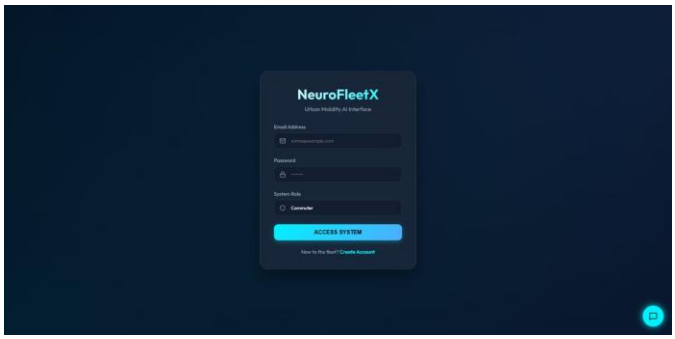

FIG 3. LOGIN INTERFACE

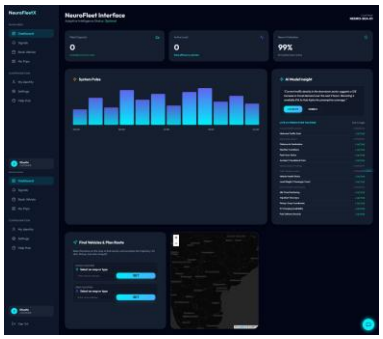

FIG 4. MAIN DASHBOARD

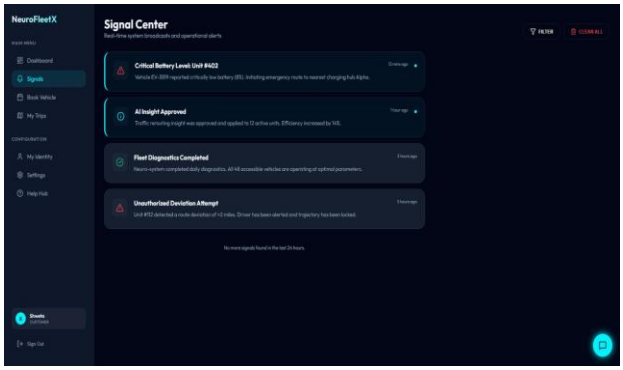

FIG 5. SIGNAL

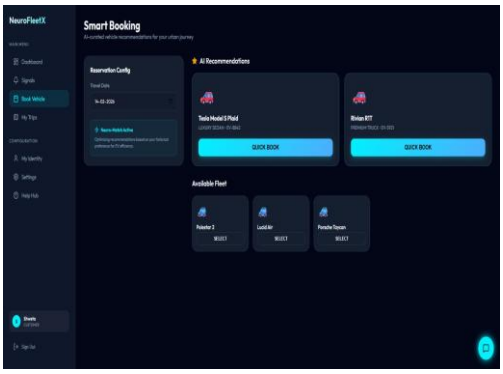

FIG 6. SMART BOOKING

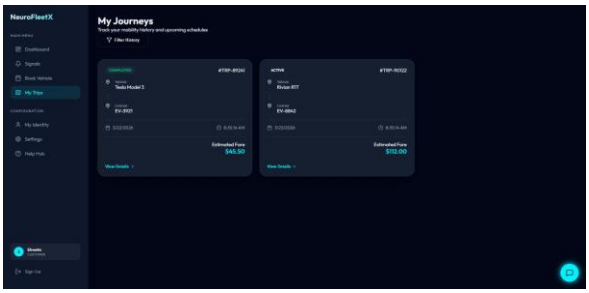

FIG 7. SIGNAL CENTER (ALERT PANEL)

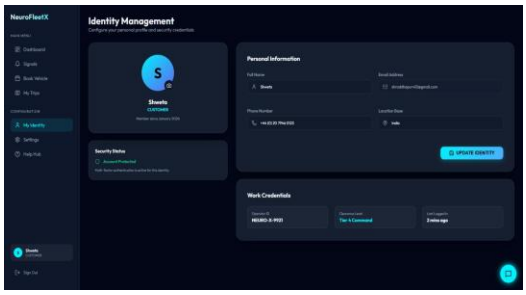

FIG 8. IDENTITY MANAGEMENT

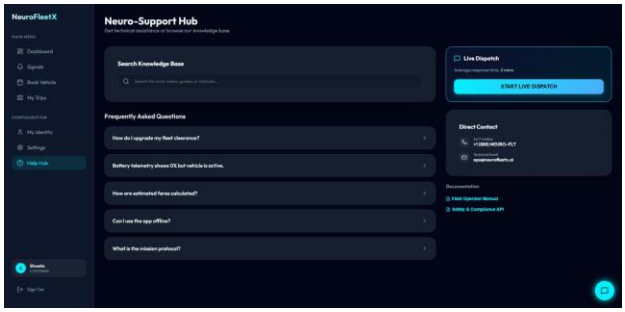

FIG 9. SUPPORT HUB

FIG 10. CHATBOT

7. Challenges Faced

- Integrating multiple modules into a single system
- Implementing AI-based logic and real-time simulation
- Managing state and navigation across pages
- Ensuring responsive UI and smooth user experience
- Debugging and optimizing performance

8. Learnings & Skills Acquired

- Full-stack development using modern technologies
- Understanding of AI-based system design
- Experience in building modular applications
- Improved debugging and problem-solving skills
- Knowledge of real-time system concepts

9. Testimonials from team

The project demonstrates strong technical skills, innovative thinking, and the ability to design and implement a complete system independently. It reflects dedication, problem-solving ability, and effective execution.

10. Conclusion

NeuroFleetX successfully presents an intelligent urban mobility solution that improves transportation efficiency through smart features such as AI-based routing, real-time tracking, and advanced dashboards. The system is scalable and can be extended for real-world applications in smart cities.

11. Acknowledgements

I would like to express my sincere gratitude to my mentor for their continuous guidance, support, and valuable feedback throughout the project. Their support played a key role in the successful completion of this work.